HDOC DAY-51

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BATCH-15

Problem Statement

Write a C program to read an integer array and pass it to a function next_greater() using pointers. Find the nearest greater element to the right of each element. Print -1 if no greater element exists.

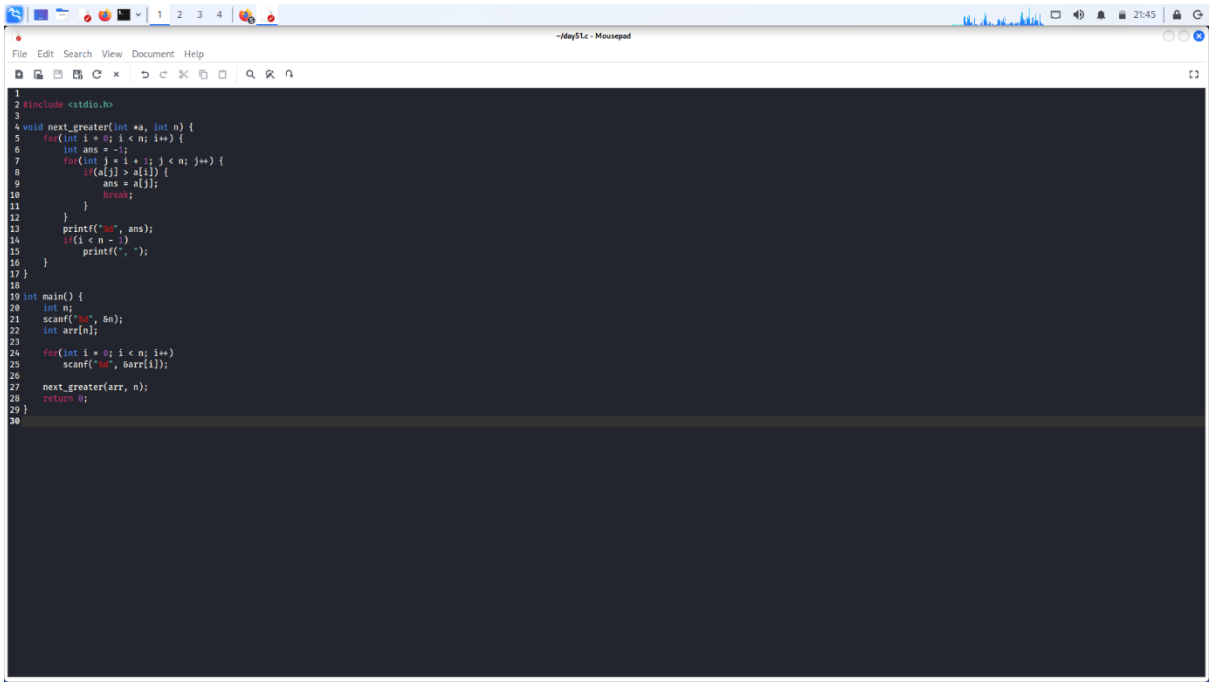
Aim

To write a C program to find the next greater element of each array element using pointers, functions, and nested loops.

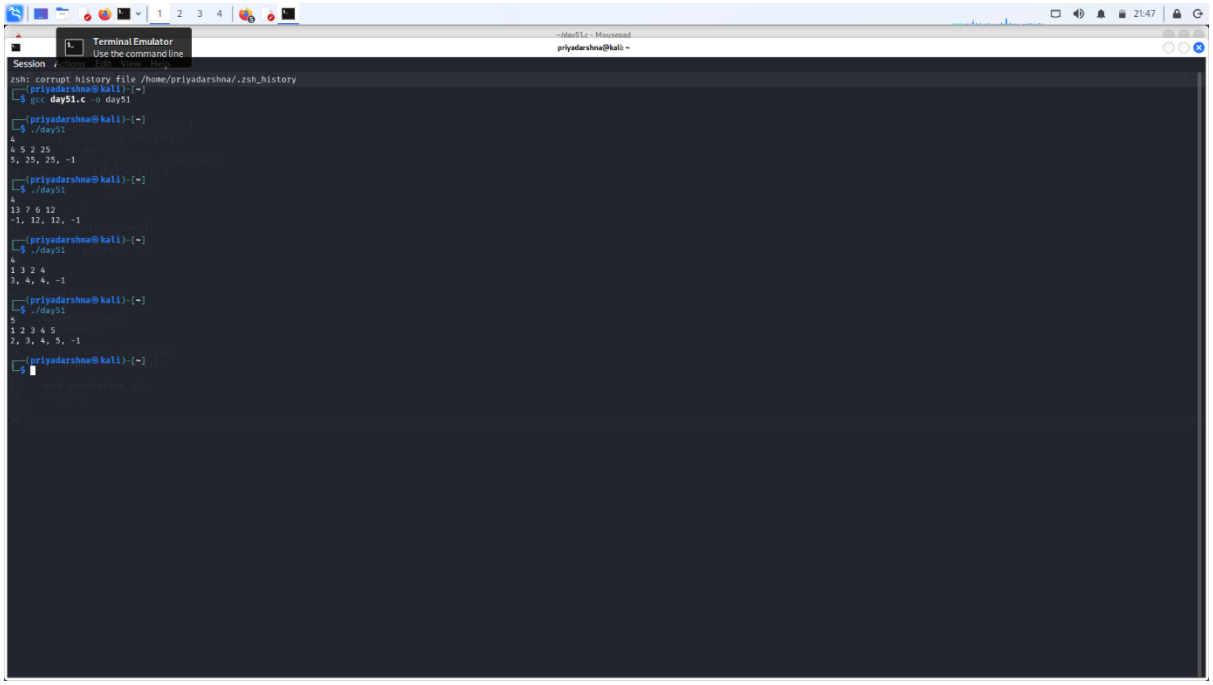
Algorithm

1. Start.
2. Read the size n of the array.
3. Read n array elements.
4. Pass the array to next_greater() using a pointer.
5. Use nested loops to check the elements to the right.
6. Print the first greater element found.
7. If no greater element exists, print -1.
8. Stop.

>C CODE


~/day51.c - Mousepad
File Edit Search View Document Help
#include <stdio.h>
void next_greater(int *a, int n) {
for(int i = 0; i < n; i++) {
int ans = -1;
for(int j = i + 1; j < n; j++) {
if(a[j] > a[i]) {
ans = a[j];
break;
}
}
printf("%d", ans);
if(i < n - 1)
printf(", ");
}
}
int main() {
int n;
scanf("%d", &n);
int arr[n];
for(int i = 0; i < n; i++)
scanf("%d", &arr[i]);
next_greater(arr, n);
return 0;
}

>OUTPUT


Terminal Emulator
Use the command line
priyadarshna@kali: ~
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna㉿kali)-[~]
$ gcc day51.c -o day51
(priyadarshna㉿kali)-[~]
$ ./day51
4
4 5 2 25
5, 25, 25, -1
(priyadarshna㉿kali)-[~]
$ ./day51
4
13 7 6 12
-1, 12, 12, -1
(priyadarshna㉿kali)-[~]
$ ./day51
4
1 3 2 4
3, 4, 4, -1
(priyadarshna㉿kali)-[~]
$ ./day51
5
1 2 3 4 5
2, 3, 4, 5, -1
(priyadarshna㉿kali)-[~]
$